Project Report

1. PROJECT TITLE

Gruha Alankara: Interior Design Platform with AI Design Agent and Augmented Reality Visualization

2. PROJECT DESCRIPTION

● System Overview

- Gruha Alankara is an advanced interior design application.
- Uses **Agentic AI and Augmented Reality**.
- Provides **personalized and multilingual design suggestions**.
- Supports **Marathi, Hindi, Telugu, and English**.
- Includes **AI agent "Buddy"** that analyzes uploaded room photos.

● Problem Domain

- People cannot visualize interior designs before implementation.
- Professional interior designers are expensive.
- The platform provides **AI-based automated design advice**.

● Core Technology Used

The system integrates multiple modern technologies to provide an interactive design experience.

- React.js and Tailwind CSS for frontend interface
- FastAPI (Python) for backend processing
- Groq Llama 3.3 model for AI design recommendations
- IBM Watsonx Vision AI for room image analysis
- gTTS for multilingual voice responses
- Three.js and WebRTC for augmented reality visualization
- SQLite database with SQLAlchemy ORM

● Key Capabilities

- AI-powered design recommendations
- Multilingual voice-based design guidance
- Intelligent room analysis using Vision AI
- AR-based furniture placement visualization
- Virtual design studio for viewing suggestions and layouts

● Target Users

- Homeowners
- Interior designers
- Real estate companies
- Students learning interior design

● Real-world Relevance

- Helps users visualize room designs before implementation.
- Reduces cost of hiring professional designers.
- Useful for **Indian users with local language support**.

3. APPLICATION SCENARIOS / USE CASES

● Enterprise Usage

Interior design companies can use the platform to present design ideas to clients through AI-generated layouts and AR visualization.

● End-User Usage

Homeowners can upload room images and receive instant design suggestions from the AI assistant.

● Educational Usage

Students studying interior design can use the platform to experiment with different layouts and color schemes.

● Industrial Usage

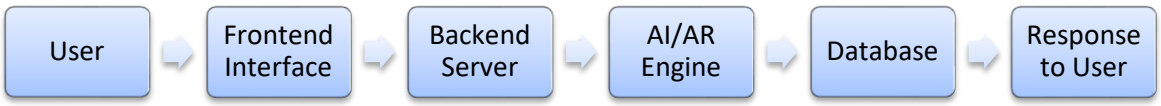
Furniture companies can showcase their products through AR-based visualization tools.

4. TECHNICAL ARCHITECTURE OVERVIEW

● High-level Architecture Diagram

The Gruha Alankara system follows a layered architecture that consists of the **Frontend Layer, Backend Layer, AI/AR Processing Layer, and Database Layer**. The frontend provides the user interface for interaction, while the backend manages application logic and communication with the database. The AI module generates design recommendations and the AR module enables real-time visualization of interior designs.

Architecture Flow:



● Component Interaction Diagram

The system contains several components that interact with each other to provide interior design functionality.

Main Components:

- **User Interface Module** – Allows users to select rooms, furniture, colors, and layouts.
- **AR Visualization Module** – Displays virtual furniture and decorations in real-world space.
- **AI Recommendation Module** – Suggests suitable design styles, layouts, and color combinations.
- **Application Server** – Processes user requests and manages communication between modules.
- **Database** – Stores user profiles, furniture models, design templates, and preferences.

Interaction Process:

1. User selects a design option through the interface.
2. The request is sent to the backend server.
3. The AI module processes the request and suggests design options.
4. The AR module visualizes the design in the user's environment.
5. The database stores user selections and preferences.

● Data / Request Flow

The data flow describes how information moves through the system during operation.

Step-by-Step Data Flow:

1. The user logs into the system and selects a room design option.
2. The frontend application sends the request to the backend server.
3. The backend processes the request and retrieves relevant data from the database.
4. The AI module analyzes the room type and suggests suitable furniture and layouts.
5. The AR module generates a real-time visualization of the selected design.
6. The generated output is sent back to the frontend and displayed to the user.
7. The system stores user preferences and design selections for future use.

5. PREREQUISITES

5.1 Software Requirements

The development and execution of the Gruha Alankara platform require the following software tools and environments.

● Development Tools

- React.js
- FastAPI
- Tailwind CSS
- Python
- Node.js
- GitHub

● IDE

- Visual Studio Code (VS Code) for frontend and backend development

● Runtime Environment

- Node.js runtime environment for backend execution
- Web browser environment for running the frontend application

5.2 Libraries / Frameworks / Dependencies

The following libraries and frameworks are required for the implementation of the system:

- **Framer Motion** – Used to create smooth animations and interactive transitions in the user interface.

- **Three.js** – A JavaScript library used for rendering and displaying 3D furniture models in the augmented reality environment.

- **WebRTC** – Enables real-time camera access in the browser for capturing room images and supporting AR-based visualization.

- **SQLAlchemy** – A Python Object Relational Mapping (ORM) library used to interact with the SQLite database efficiently.

- **gTTS (Google Text-to-Speech)** – Converts AI-generated design suggestions into spoken voice guidance in different languages.

- **Groq API** – Provides access to the Llama 3.3 language model used for generating intelligent interior design recommendations through the AI agent “Buddy”.

- **IBM Watsonx API** – Used for Vision AI capabilities such as analyzing uploaded room images and detecting objects within the room.

5.3 Hardware Requirements

● System Specifications

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: At least 20 GB free space
- Operating System: Windows 10 / macOS / Linux

● Devices / Sensors

- Smartphone or tablet with camera support for AR functionality
- Webcam or mobile camera for real-time room visualization
- Internet connection for accessing cloud services and databases

6. PRIOR KNOWLEDGE REQUIRED

Before implementing the Gruha Alankara platform, developers should have basic knowledge of several technical concepts and tools. This prior knowledge helps in understanding the system architecture, development process, and integration of advanced technologies such as Artificial Intelligence and Augmented Reality.

● Programming Language Knowledge

Developers should have knowledge of programming languages used in the project such as **JavaScript, HTML, and CSS** for frontend development and **Node.js or Python** for backend logic. Understanding basic programming concepts such as variables, functions, loops, and object-oriented programming is important for building the system.

● Framework Basics

Knowledge of web development frameworks such as **React.js** for frontend and **Express.js** for backend development is helpful. These frameworks simplify the development process by providing structured ways to build user interfaces and manage server-side operations.

● Database Fundamentals

Basic understanding of databases such as **MongoDB or Firebase** is required. Developers should know how to store, retrieve, update, and manage data related to user profiles, interior design templates, furniture models, and user preferences.

● Networking / Cloud Concepts

Understanding basic networking concepts such as client-server communication, APIs, and cloud hosting platforms is useful. Knowledge of cloud services helps in deploying the application and managing online data storage.

● AI / ML Fundamentals

Since the system uses Artificial Intelligence for design recommendations, basic knowledge of AI and Machine Learning concepts such as data processing, model training, and prediction is beneficial. These concepts help in generating intelligent suggestions for interior layouts, color schemes, and furniture placement.

7. PROJECT OBJECTIVES

The main objective of the Gruha Alankara project is to develop an intelligent interior design platform that helps users visualize and plan their home interiors using modern technologies such as Artificial Intelligence and Augmented Reality.

● Technical Objectives

- To design and develop an interactive platform for interior design visualization.
- To integrate **Augmented Reality (AR)** for real-time visualization of furniture and room layouts.
- To implement **Artificial Intelligence (AI)** for providing smart design recommendations.
- To create a user-friendly interface that allows easy interaction and customization of room designs.

● Performance Objectives

- To provide fast and responsive system performance for real-time visualization.
- To ensure accurate placement of virtual furniture and interior elements using AR technology.
- To optimize system performance for smooth interaction and minimal loading time.
- To maintain reliable data storage and retrieval for user designs and preferences.

● Deployment Objectives

- To deploy the system as a web or mobile application that can be accessed easily by users.
- To ensure the platform can run on standard devices such as smartphones, tablets, and computers.
- To support integration with cloud services for data storage and system scalability.

● Learning Outcomes

- To gain practical experience in developing real-world software applications.
- To understand the integration of **Artificial Intelligence and Augmented Reality** in modern applications.
- To improve knowledge of full-stack development including frontend, backend, and database technologies.
- To develop problem-solving, teamwork, and project management skills.

8. SYSTEM WORKFLOW

The system workflow describes the step-by-step process of how the Gruha Alankara platform operates from user interaction to the final output delivery.

1. User Interaction

The user accesses the Gruha Alankara platform through a web or mobile interface. The user logs into the system and selects options such as room type, furniture style, color schemes, or interior design preferences.

2. Input Handling

The system collects the user's inputs such as room dimensions, selected furniture items, and design preferences. These inputs are validated and processed by the frontend before being sent to the backend server.

3. Processing Logic

The backend server processes the received data and communicates with the AI module. The AI system analyzes the inputs and suggests suitable furniture layouts, color combinations, and interior design styles based on predefined models and design rules.

4. Core System Execution

The Augmented Reality module generates a virtual representation of the selected furniture and design elements. The system places these elements in the user's real environment through AR visualization.

5. Output Generation

The system generates a visual preview of the designed room. The user can see how furniture, colors, and decorations appear together in a realistic environment.

6. Response Delivery

The final visualization and design suggestions are displayed to the user through the application interface. Users can modify the design, save their layouts, or explore different design options.

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

9.1 Problem Definition

Interior design planning is often difficult for homeowners because they cannot visualize how furniture, colors, and decorations will look in their living spaces before implementation. Traditional interior design services can be expensive and time-consuming. There is a need for a digital platform that allows users to experiment with interior designs and view realistic results before making final decisions.

The Gruha Alankara platform aims to solve this problem by providing an interactive system that uses **Artificial Intelligence (AI)** to recommend suitable designs and **Augmented Reality (AR)** to visualize furniture and layouts in real environments.

9.2 Functional Requirements

Functional requirements describe the main features that the system must perform.

- User registration and login functionality
- Selection of room type and interior design preferences
- Uploading or scanning room space for visualization
- AI-based suggestions for furniture arrangement and color combinations
- Multilingual voice support
- AR visualization of furniture and interior elements in the real environment
- Ability to modify and customize interior layouts
- Saving and managing user design projects

9.3 Non-Functional Requirements

Non-functional requirements describe system performance and quality attributes.

- **Usability:** The system should provide a simple and user-friendly interface.
- **Performance:** The platform should provide quick response time for design generation and visualization.
- **Reliability:** The system should operate without frequent crashes or failures.
- **Scalability:** The platform should support multiple users simultaneously.
- **Security:** User data and design preferences should be securely stored and protected.

9.4 System Design Decisions

Several design decisions were made to ensure efficient system development and performance.

- A **layered architecture** was selected to separate frontend, backend, AI, and AR modules.
- The system uses a **web-based interface** so that users can easily access it through browsers.
- AI modules are integrated to generate intelligent design suggestions.

- AR technology is used to create realistic visualization of interior designs in real space.
- A database is used to store user information, furniture models, and design templates.

9.5 Technology Stack Selection Justification

The technology stack was selected based on performance, scalability, and ease of development.

- **Frontend:** HTML, CSS, JavaScript, and React.js are used to build a responsive and interactive user interface.
- **Backend:** Node.js and Express.js are used for server-side processing and API management.
- **Database:** MongoDB or Firebase is used to store user data, design templates, and furniture models.
- **AI Technology:** Machine learning libraries are used to generate interior design recommendations.
- **AR Technology:** AR libraries such as WebXR or AR.js are used to display furniture in the user's real environment.

Diagrams to Include in the Report

● Architecture Diagram

Draw a diagram showing:

User → Frontend → Backend Server → AI Module → AR Module → Database

● ER Diagram (Example Entities)

You can include entities such as:

- User
- Design Project
- Furniture Item
- Room Layout

● API Structure (Example)

Example API endpoints:

- **POST /login** – User authentication
- **POST /register** – User registration
- **GET /designs** – Fetch saved designs
- **POST /generate-design** – Generate AI-based design suggestions
- **POST /save-design** – Save user design project

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

10.1 Development Environment Setup

The development environment was prepared to support frontend, backend, AI, and AR integration. Visual Studio Code was used as the main development environment for coding and debugging. Git and GitHub were used for version control and collaboration among team members. Web browsers such as Google Chrome were used for testing the application.

10.2 Dependency Installation

Various libraries and frameworks were installed to support system development.

Example setup commands:

- Install Node.js ,pip and npm
- Install project dependencies using command:
  npm install
- Install additional packages for AR and AI modules as required.

10.3 Project Structure Creation

The project was organized into different folders to maintain clean and modular code structure. This helps developers manage frontend, backend, and configuration files separately.

Example Folder Structure

```
gruha-alankara/
│
├── frontend/ (React + Tailwind UI)
├── backend/ (FastAPI server)
├── ai_agent/ (Groq AI integration)
├── vision_ai/ (IBM Watsonx models)
├── ar_engine/ (Three.js AR rendering)
├── database/
├── assets/
└── main.py
```

10.4 Configuration Setup

Configuration files were created to manage environment variables, database connections, and application settings.

Examples include:

- Database connection configuration
- API keys for AI or AR services

- Environment variables for server configuration

This setup ensures smooth communication between system components.

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

11.1 Feature 1 – Multilingual AI Design Buddy

Description

- AI agent **Buddy** analyzes uploaded room images.
- Generates **color palettes, furniture suggestions, lighting advice**.

Code Explanation

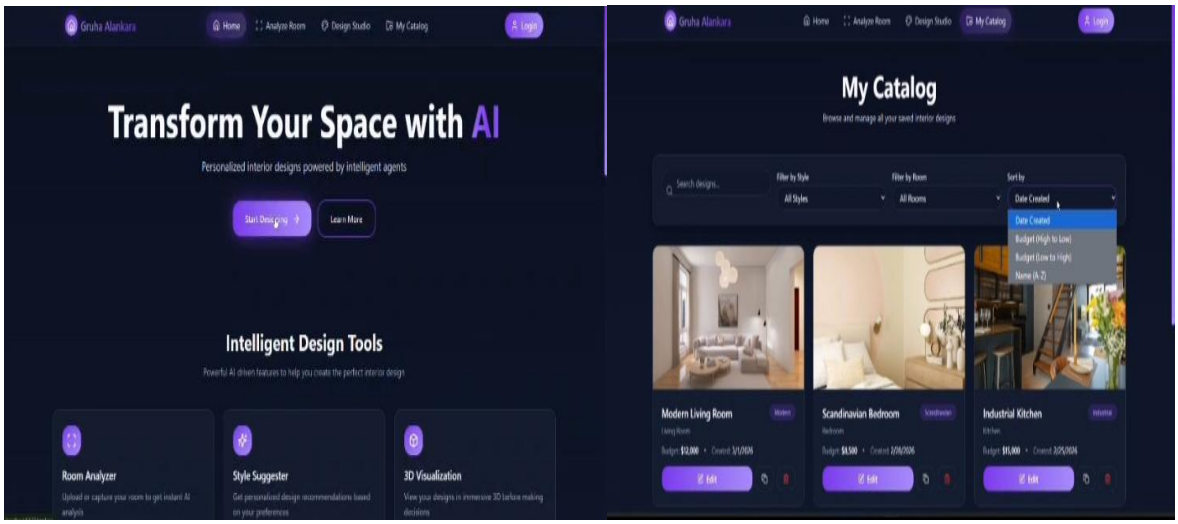
The Multilingual AI Design Buddy is implemented using a backend API built with **FastAPI (Python)**. When a user uploads a room image, the frontend sends the image to the backend server through an HTTP request. The backend processes the image and sends it to the **Groq API** which runs the Llama 3.3 language model.

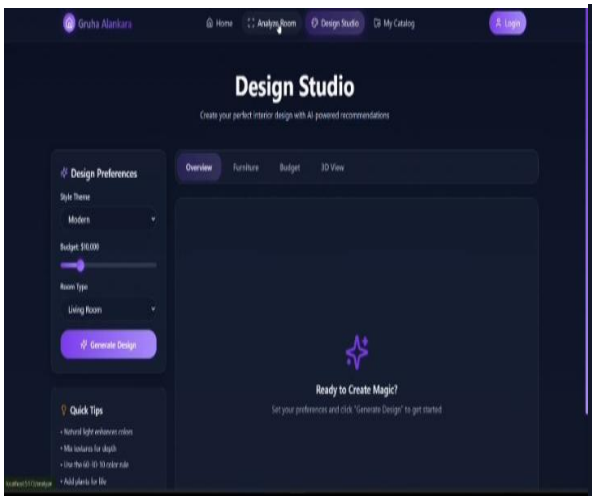
The AI model analyzes the room layout and generates design suggestions such as color palettes, furniture arrangements, and lighting recommendations. These suggestions are returned in JSON format.

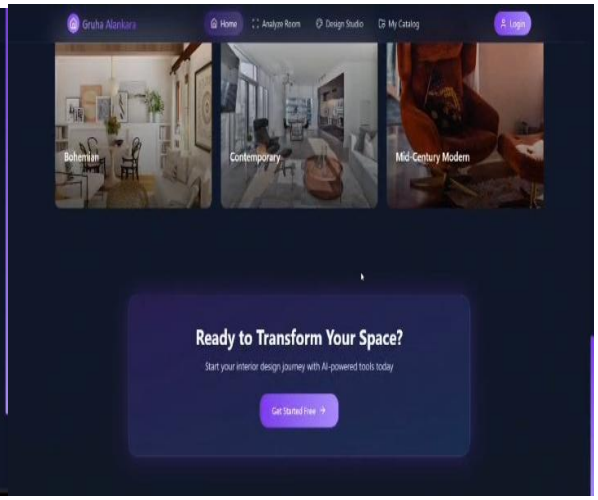
To provide voice guidance in the user's selected language, the system uses **gTTS (Google Text-to-Speech)**. The generated design suggestions are converted into spoken audio and played in the user interface.

The backend also stores the generated suggestions and image paths in the database using **SQLAlchemy**, which allows users to revisit their design history later.

Screenshots / Outputs







11.2 Feature 2 – Intelligent Room Analysis

Description

- Vision AI detects objects in the room.
- Suggests **Modern, Classic, Industrial, or Minimalist themes**.

Implementation

The Intelligent Room Analysis feature uses **IBM Watsonx Vision AI** to analyze uploaded room images. After the image is uploaded, the backend API sends the image data to the Vision AI service.

The Vision AI model detects objects such as walls, windows, furniture, and available space in the room. Based on the detected elements and spatial layout, the system determines suitable interior design styles such as Modern, Classic, Industrial, or Minimalist.

The analysis results are returned to the backend in structured data format. The backend then processes this data and sends design suggestions to the frontend interface where users can view recommended themes and design ideas.

This process allows the system to provide intelligent and context-aware design recommendations.

11.3 Feature 3 – AR Furniture Placement

Description:

- WebRTC camera access.
- Three.js for placing **3D furniture models in the room**.

Implementation

The AR Furniture Placement feature allows users to visualize virtual furniture inside their real environment. The system uses **WebRTC** to access the device camera directly through the browser.

Once the camera feed is active, the application uses **Three.js**, a JavaScript library for rendering 3D graphics, to display virtual furniture models on top of the live camera view.

The user can select furniture items from the interface, and the application loads corresponding 3D models. These models are positioned and scaled within the scene to match the room environment.

The rendering engine continuously updates the camera view and 3D objects in real time, allowing users to move their device and see how furniture appears from different angles. This helps users make better interior design decisions before purchasing or arranging furniture.

Database Schema (Example)

Main database entities include:

- **User**
- **Design Project**
- **Furniture Item**
- **Room Layout**

These entities store user information, design preferences, and furniture details.

Business Logic Layer

The business logic layer processes user inputs, communicates with the AI recommendation module, and sends appropriate responses to the frontend interface.

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

12.1 Component Integration

All system components including the frontend interface, backend server, database, AI module, and AR visualization module were integrated to ensure smooth communication and functionality.

12.2 Performance Optimization

Performance optimization techniques were applied to improve application speed and responsiveness. Efficient database queries and optimized rendering of 3D objects were used to reduce loading time.

12.3 Security Enhancements

Security mechanisms such as user authentication, password encryption, and secure API communication were implemented to protect user data and system integrity.

12.4 Error Handling & Validation

The system includes proper error handling mechanisms to detect invalid inputs, server errors, or system failures. Input validation ensures that only valid data is processed by the system.

13. MILESTONE 5: TESTING & VALIDATION

13.1 Test Cases

Test Case ID	Input	Expected Output	Status
TC01	User login credentials	Successful login	Passed
TC02	Invalid login credentials	Error message displayed	Passed
TC03	Room design selection	Design suggestions generated	Passed
TC04	AR visualization request	Furniture displayed in AR view	Passed

13.2 Unit Testing

Unit testing was performed on individual modules such as authentication, AI recommendation engine, and AR visualization module to ensure that each component works correctly.

13.3 Integration Testing

Integration testing verified that different modules such as frontend, backend, and database interact properly without errors.

13.4 User Acceptance Testing

User testing was conducted to ensure the system meets user requirements and provides a satisfactory design experience.

13.5 Performance Metrics

For web or application performance evaluation:

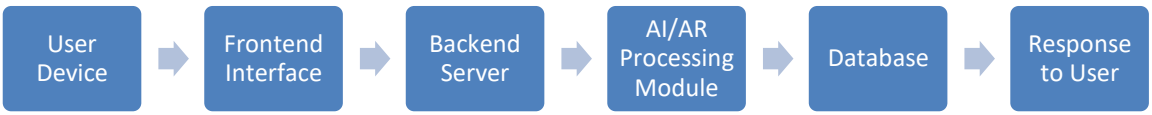
- **Response Time:** The system provides quick response time for user requests.
- **Load Handling:** The platform can handle multiple user requests simultaneously.
- **Security Validation:** Authentication and data protection mechanisms ensure secure operation of the system.

14. DEPLOYMENT

14.1 Deployment Architecture

The deployment architecture of the Gruha Alankara platform follows a client–server model. The frontend application runs in the user's web browser or mobile device, while the backend server processes requests and communicates with the database. The AI module generates design recommendations and the AR module renders virtual furniture models for visualization.

Architecture flow:



14.2 Hosting Platform

The platform can be deployed using modern hosting services such as:

- GitHub for source code management
- Netlify or Vercel for frontend hosting
- Node.js server hosting platforms for backend deployment
- Cloud database services such as MongoDB Atlas or Firebase

These platforms allow easy deployment, scalability, and remote access to the system.

14.3 Deployment Steps

1. Upload the project code to the GitHub repository.
2. Configure the backend server and database connection.
3. Deploy the frontend application on a hosting platform such as Netlify or Vercel.
4. Connect the frontend application with backend APIs.
5. Test the deployed system to ensure proper functionality.

14.4 Production Considerations

Before deploying the system in production, several factors must be considered:

- Secure handling of user data and authentication.
- Proper database backup and recovery mechanisms.

- Performance optimization for handling multiple users.
- Regular system updates and maintenance.

15. PROJECT STRUCTURE

Folder Structure

```
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├── database/
├── assets/
└── main.py
```

16. RESULTS

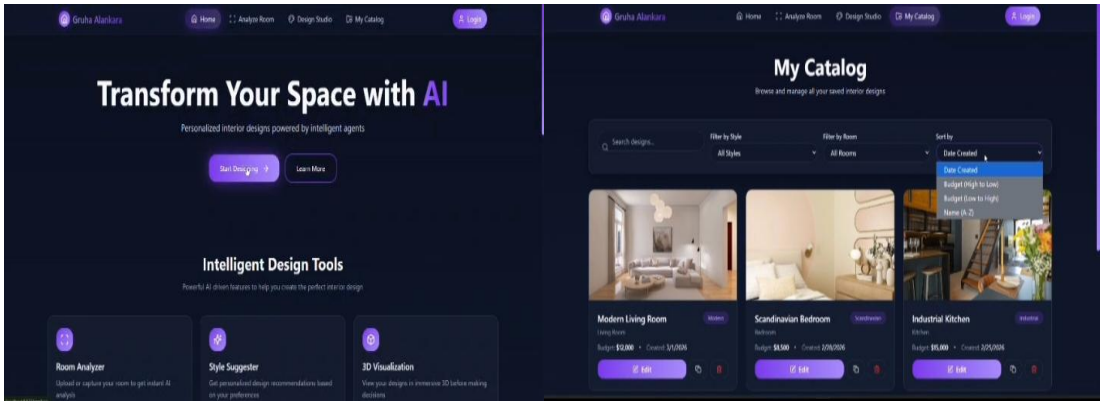
System Output

- Project completion: **90%**
- AI integrations working (Groq, Watsonx, gTTS)
- AR placement logic under final testing.

Performance Evaluation

The system provides smooth interaction and fast response time for design generation and visualization. The AR module allows realistic placement of furniture in the user's environment.

Screenshots



Benchmark Results

Testing results show that the system performs efficiently for normal user interactions and can handle multiple design operations with minimal delay.

17. ADVANTAGES & LIMITATIONS

Advantages

- Provides realistic visualization of interior designs using AR technology.
- AI-based recommendations help users choose suitable designs.
- Reduces the cost and effort of traditional interior design planning.
- Easy-to-use interface for both professionals and beginners.
- Helps users avoid design mistakes before actual implementation.

Limitations

- AR visualization may require modern devices with camera support.
- Accuracy of AI recommendations depends on training data.
- High-quality 3D models may require more processing power.
- Internet connection may be required for cloud-based services.

18. FUTURE ENHANCEMENTS

Several improvements can be implemented in future versions of the system:

Scalability Improvements

The system can be optimized to support a larger number of users and more complex interior design models.

Feature Expansion

Additional features such as lighting simulation, budget estimation, and furniture shopping integration can be added.

Cloud Integration

The platform can be integrated with cloud services to store user designs and provide collaborative design options.

Mobile Integration

A dedicated mobile application can be developed to improve user experience and accessibility.

Automation

Advanced AI models can automatically generate complete room designs based on minimal user input.

19. CONCLUSION

The Gruha Alankara platform successfully demonstrates the integration of Artificial Intelligence and Augmented Reality for interior design visualization. The system provides an interactive and user-friendly environment where users can experiment with different design ideas and view realistic results before implementation.

Through this project, practical knowledge of web development, AI integration, and AR visualization was gained. The platform has the potential to simplify interior design planning and improve decision-making for homeowners, designers, and businesses.

20. APPENDIX

The appendix contains supporting materials and additional resources related to the project.

Source Code

The complete source code of the **Gruha Alankara AI Interior Design Platform** is available in the GitHub repository. The project includes both frontend and backend modules that work together to provide AI-powered interior design suggestions.

The frontend is developed using **HTML, CSS, JavaScript, and React.js**, which provides an interactive user interface where users can upload room images, explore design suggestions, and view interior layouts.

The backend is developed using **Python and FastAPI**, which processes user requests, communicates with AI services, and manages application logic. The backend also handles image processing, AI model communication, and system responses.

The repository also contains configuration files, documentation, and supporting resources required to run the application.

Configuration Files

Configuration files are used to set up the system environment and connect different components of the application.

These files include:

- Environment configuration files used to store API keys and environment variables.
- Database configuration files used to establish connection with the database.
- Backend server configuration for running the FastAPI application.
- Frontend dependency files used for installing required libraries and packages.

These configuration files ensure that the application can be properly deployed and executed on local machines or cloud platforms.

API Documentation

The system uses APIs to enable communication between the frontend interface and the backend services.

The backend provides REST API endpoints that perform the following operations:

- **User Authentication API** – Handles user registration and login.
- **Image Upload API** – Allows users to upload room images.
- **AI Processing API** – Sends images to AI services for object detection and design generation.
- **Design Recommendation API** – Returns AI-generated interior design suggestions.
- **AR Visualization API** – Enables augmented reality visualization of furniture placement.

These APIs ensure smooth interaction between different components of the system and allow the platform to provide real-time design recommendations.

GitHub Repository Link

https://github.com/Swapnil-Ilake2608/Gruha-Alankara-Interior-Design-Platform-with-AR-and-AI-Integration

Demo video link

https://drive.google.com/drive/folders/1M8hbb7tkYIcttUK_7ENsNDHvGuSK8HzF?usp=sharing

Live Demo Link

https://drive.google.com/drive/folders/1M8hbb7tkYIcttUK_7ENsNDHvGuSK8HzF?usp=sharing